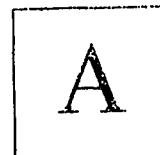


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B.Tech. Degree VII Semester Regular/Supplementary Examination November 2023

**ME 19-205-0706 AUTOMOBILE ENGINEERING
(2019 Scheme)**

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Compare the working of various types of automobiles.
 CO2: Learn the functions and working of different engine components.
 CO3: Appreciate the use of alternate fuels.
 CO4: Understand the latest technology used in the field of automobile.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate,
 L6 – Create
 PO – Programme Outcome

PART A
(Answer *ALL* questions)

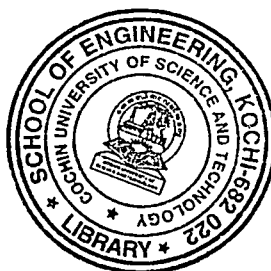
	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a) Discuss combustion chamber types.	3	3	L2	2	
(b) What do you mean by 'valve clearance'?	3	3	L2	3	
(c) Name the main components of fuel supply system for automotive diesel engines with the help of a neat diagram	3	3	L2	2	
(d) What are the advantages of using CNG as an alternate fuel?	3	3	L2	3	
(e) What are the functions of transmission in a vehicle?	3	3	L2	2	
(f) What are the advantages of constant mesh gear box when compared with sliding mesh type gear box?	3	3	L2	2	
(g) Mention the advantages of hybrid vehicles.	3	3	L2	4	
(h) Draw the basic block diagram of electric vehicle showing its components and explain.	3	3	L2	4	

PART B

(4 × 12 = 48)

II. (a) Define the term Piston slap. What is the cause of piston slap?	4	4	L2	1	
(b) Discuss in detail about the different materials used to manufacture pistons, cylinders, connecting rods and crankshaft.	8	8	L2	2	
OR					
III. (a) Explain VTEC technology to effect variable valve timing in automobiles with a neat sketch.	6	6	L2	4	
(b) Discuss ignition advance and its features.	6	6	L2	2	

(P.T.O.)



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		Marks	BL	CO	PO
IV.	(a) What are the different types of fuel supply systems for a petrol engine?	2	L2	2	
	(b) What is the necessity of a cooling system? Explain a cooling system with a neat sketch.	10	L3	2	
OR					
V.	(a) What are the requirements of an ignition system of an internal combustion engine?	2	L3	2	
	(b) Discuss in detail about the battery ignition system. Illustrate your answer with neat sketches.	10	L2	2	
VI.	(a) Explain drive train of an automobile.	2	L2	1	
	(b) Explain the necessity of a differential in an automobile. Discuss in detail the operation of a differential with a neat sketch.	10	L2	2	
OR					
VII.	(a) What is a CVT? Discuss its main advantages and limitations.	5	L2	4	
	(b) Explain the working of synchromesh gear box with neat sketches.	7	L2	2	
VIII.	(a) Explain the concept of electric traction.	6	L2	4	
	(b) Discuss in detail about Electric Vehicle Drive train alternatives based on Power source configuration.	6	L2	4	
OR					
IX.	(a) Discuss in detail about energy storage systems in hybrid electric vehicles.	8	L2	4	
	(b) What do you mean by high energy density and high power density? Explain with examples.	4	L2	4	

Bloom's Taxonomy Levels
L2 = 90%, L3 = 10%.
